

ORIGINAL ARTICLE

# Subinhibitory concentrations of nalidixic acid alter bacterial physiology and induce anthropogenic resistance in a commensal strain of *Escherichia coli in vitro*

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**Significance and Impact of the Study:** The effect of sublethal concentrations of antibiotics on commensal micro-organisms is less studied compared to that of pathogenic micro-organisms. For the first time, the influence of nalidixic acid (NA) sub-MICs on a commensal strain of *E. coli* was reported. The findings precisely demarcate the role of NA sub-MICs in altering bacterial growth rate and physiology and inducing genomic aberrations, thus resulting in the emergence of antibiotic resistance. This justifies the implications of sublethal concentrations of antibiotics as signalling molecules that drive microbial pathogenicity and disseminate antimicrobial resistance (AMR).

## Keywords

commensal *Escherichia coli*, induced antibiotic resistance, nalidixic acid, OMP expression, QRDRs, sub-MIC.

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## Abstract

The human gut houses a complex group of bacterial genera, including both opportunistic pathogens and commensal micro-organisms. These are regularly exposed to antibiotics, and their subinhibitory concentrations play a pivotal role in shaping the microbial responses. This study was aimed to investigate the effects exerted by sub-MICs of nalidixic acid (NA) on the growth rate, bacterial motility, biofilm formation and expression of outer membrane proteins (OMPs) in a commensal strain of *E. coli*. The NA-sensitive strain was sequentially passaged under sub-MICs of NA. E-test was used to determine the MIC values of NA. Results indicated significant changes in the growth profile of commensal *E. coli* upon exposure to NA at sub-MICs. Differential expression of OMPs was observed in cells treated with sub-MICs of NA. Bacterial motility was reduced under 1/2 MIC of NA. Interestingly, successive passaging under 1/2 MIC of NA led to the emergence of resistant *E. coli* with an increased MIC value of 64 µg ml<sup>-1</sup> in just 24 days. The NA-resistant variant was confirmed by comparing its 16S rRNA sequence to that of the sensitive commensal strain. Mutations in the Quinolone Resistance-Determining Regions (QRDRs) of chromosomal *gyrA*, and Topoisomerase IV-encoding *parC* genes were detected in NA-resistant *E. coli*. Our results demonstrate how antibiotics play an important role as signalling molecules or elicitors in driving the pathogenicity of commensal bacteria *in vitro*.

## Introduction

*Escherichia coli* is an integral part of the in-house microflora and constitutes approximately 0.1% of the gut microbiome (Eckburg *et al.* 2005). Most *E. coli* exist as commensals regulating intestinal harmony by holistically co-existing with their microbial counterparts and promoting numerous biological functions, including metabolism, immunomodulation, secretion of growth factors and

vitamins (Yang and Jobin 2014). Based on the presence of diverse genomic elements and interactions with the host, *E. coli* has been categorized into seven phylogroups; A, B1, and C are classified as commensals, while B2, D, E, and F comprise pathogenic subtypes (Clermont *et al.* 2013). With the development of third- and fourth-generation antibiotics, many bacterial infections once considered deadly have now become curable. However, the extensive consumption of these drugs has also

accelerated the pace at which antimicrobial resistance (AMR) evolves in micro-organisms (Chadha *et al.* 2021). These ever-evolving microbial strategies have led to the emergence of multi-drug resistant (MDR) pathogens or superbugs. Today, this phenomenon is snowballing at an alarming rate and is expected to burden global economies by 100 trillion USD and 10 million deaths by 2050 (Chadha *et al.* 2021).

In developing countries with high population density and inadequate sanitary practices, the prevalence of bacterial infections is very high. As a consequence, the use of antibiotics to treat bacterial infections is on the rise. For a long time, quinolone antibiotics have been used extensively for combating a wide variety of human pathogens (Pandey *et al.* 2018). Amongst the first chemically synthesized quinolones, nalidixic acid (NA) has been commonly prescribed as a generic medicine in 42% of urinary tract infections worldwide, especially in developing countries like India, Bangladesh, Kenya, Uganda and Pakistan (Laxminarayan and Chaudhury 2016; Sharma *et al.* 2017). However, due to the emergence of widespread bacterial resistance and the development of robust antimicrobials like fluoroquinolones, its application in clinical setups has been limited (Sharma *et al.* 2017). Nevertheless, NA continues to be used clinically in several countries such as Pakistan, Nigeria, Bangladesh, China, Indonesia and some parts of India (Sharma *et al.* 2017). In this regard, antibiotic administration has been shown to disrupt intestinal homeostasis, eliminating both pathogenic and commensal bacteria, and inducing dysbiosis of the gut microflora (Konstantinidis *et al.* 2020). However, the surviving commensals experience subminimum inhibitory concentrations (sub-MICs) upon antibiotic clearance from the body that exerts significant and unknown outcomes with time, including horizontal gene transfer, biofilm formation, enhanced cell adhesion and spontaneous mutations (Mlynek *et al.* 2016; Gümüş *et al.* 2018; Walsh *et al.* 2019; Chadha 2021). This may also cause the passage of resistance from commensal strains to other microbial populations in the gut horizontally. These findings have drawn interest in the effects of sub-MIC exposure to bacteria, so that pharmacodynamics and pharmacokinetic data can be correlated for the optimization of antibiotic therapy.

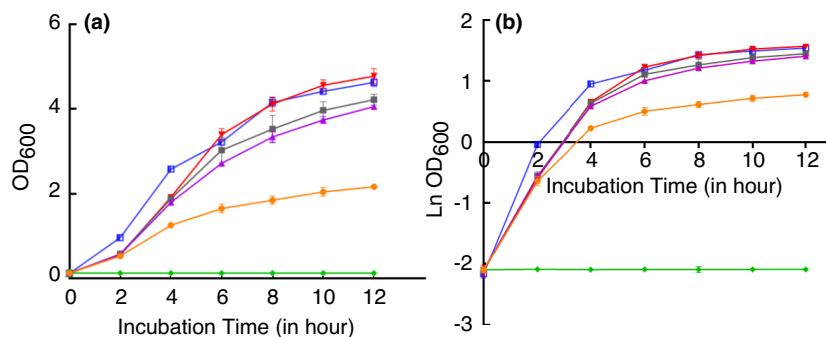
Presently, there is a dearth of knowledge regarding the effects produced by sub-MICs of antibiotics on commensal bacteria of the human gut. Nevertheless, experimental studies carried out in the last few years have demonstrated that sub-MICs produce various effects on bacterial cells. Investigations with sub-MICs of NA have shown variable findings. Exposure of *Pseudomonas aeruginosa* to sublethal concentrations of NA resulted in cell deformation with elongated morphology and differential expression of

cellular proteins (Al Bahry *et al.* 2012). NA at sub-MIC values has been reported to induce biofilm formation in clinical isolates of *Salmonella* Infantis (Tezel *et al.* 2016) while reducing the growth rate and expression of type II toxin-antitoxin system genes in *Klebsiella pneumoniae* (Narimisa *et al.* 2020). Moreover, sublethal concentrations of ciprofloxacin have been documented to induce genetic mutations in *E. coli* DNA through intra-chromosomal recombination (Lopez *et al.* 2007; Lopez and Blazquez 2009). Since there are gaps in our existing knowledge about the long-term effects of antibiotic sub-MIC exposure on commensals, such findings raise interest in getting to the core of this avenue. Hence, we formulated a hypothesis that NA at sub-MIC levels can also regulate the bacterial responses in a commensal *E. coli* strain. The objective of this study was therefore aimed at scrutinizing the intriguing effects produced in a commensal strain of *E. coli* isolated from River Yamuna, New Delhi, following exposure to NA at sub-MIC levels. The impact on bacterial growth kinetics, motility, biofilm formation, expression of outer membrane proteins (OMPs) and quinolone resistance-determining regions (QRDRs) elements of gyrase and topoisomerase IV genes were tested *in vitro*. Our investigation points towards the possible role of NA in governing microbial responses that may lead to the development of antibiotic resistance among commensal microflora.

## Results and discussion

### NA at sub-MIC level reduces the bacterial growth rate

Antibiotics at sublethal concentrations have been demonstrated to alter the growth kinetics of several bacterial species (Reeks *et al.* 2005; Chadha 2021). However, very little is known in this regard about commensal microorganisms. To gain insights into this, the growth pattern of a commensal *E. coli* strain was studied following exposure to NA sub-MICs for 7 days at 37°C. Since NA is an orally administered drug prescribed for 7 days (Hovelius *et al.* 1985), we passaged the commensal strain under NA sub-MICs for the same duration. During the period of bacterial growth at 37°C, an ideal sigmoidal growth curve with distinct lag, log and stationary phases was observed (Fig. 1). The NA sub-MICs altered the bacterial growth rate and doubling time in a concentration-dependent manner. With NA at 1/2 MIC, the lag phase was prolonged with restricted bacterial growth. Subsequent sub-MICs (1/4 and 1/8) did not show any significant changes in the growth kinetics of the commensal strain. This effect can be justified as the antibiotic gets diluted to empirically lower concentrations that are unable to hinder or retard bacterial growth under experimental conditions



**Figure 1** The effect of sub-MICs of NA on growth kinetics of commensal *Escherichia coli* at 37°C. (a) Growth profile at OD<sub>600</sub>. (b) Ln OD<sub>600</sub> plot for determining growth parameters. Sub-MICs: (◆) At MIC, (●) 1/2 MIC, (▲) 1/4 MIC, (■) 1/8 MIC; (▼) AFC and (■) ATCC 25922.

(Chadha 2021). In the presence of NA at 1/2 MIC, the doubling time increased significantly to  $72.53 \pm 1.187$  min in comparison to antibiotic-free culture (AFC) ( $60.17 \pm 1.899$  min). Meanwhile, the quality control strain, *E. coli* ATCC 25922, showed prolific growth with a doubling time of  $39.15 \pm 0.748$  min and a growth rate of  $0.652 \pm 0.012$  h<sup>-1</sup>. The effect on growth kinetics upon exposure to NA sub-MICs has been summarized in Table S2. Our findings are in line with previous studies that demonstrate the effect of antibiotic sub-MICs on bacterial growth kinetics (Gümüş *et al.* 2018; Waack and Nicholson 2018). Since NA inhibits the activity of DNA gyrase and Topoisomerase IV (Hooper and Jacoby 2016), it is logical that the process of DNA replication would gradually undergo retardation, causing slow bacterial growth in the presence of NA at sub-MIC levels. This suggests that NA at 1/2 MIC significantly influences the growth kinetics of the commensal *E. coli* strain.

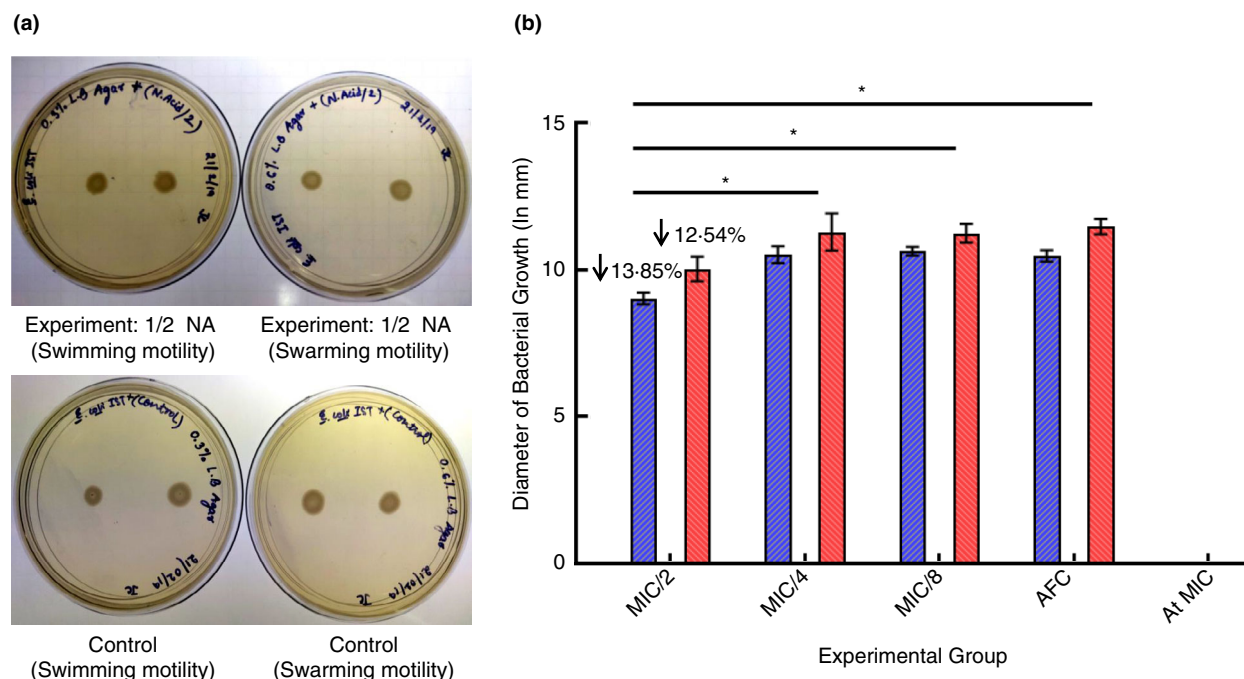
### Sub-MIC of NA retards bacterial motility

Recent studies have confirmed that antibiotics at sublethal concentrations can have diverse but extreme effects on bacterial motility. Subinhibitory levels of arylomycin have been shown to strongly inhibit both swimming and swarming motility in *E. coli*, thereby impeding biofilm formation (Walsh *et al.* 2019). Interestingly, studies have also reported antibiotic sub-MICs to enhance bacterial motility (Shi *et al.* 2019). Discrete findings have also pinpointed the inhibition of bacterial motility *via* attenuation of key virulence elements (Gupta *et al.* 2016; Walsh *et al.* 2019). Motility in bacteria is known to play a crucial role in the establishment of infection during the initial phases as it aids in colonization, especially for pathogens and gut commensals (Haiko and Westerlund-Wikstrom 2013). Therefore, the effect of NA sub-MICs on bacterial motility, particularly swimming and swarming motility of commensal *E. coli* strain, were investigated using standard

plate assays. The diameters of bacterial growth were analysed by a one-way analysis of variance (ANOVA) test which revealed a significant reduction in swimming and swarming motility under 1/2 MIC (Fig. 2). NA at 1/4 and 1/8 MICs did not manifest any consequential effects on the motility of the commensal *E. coli*. Previously, studies by Wojnicz *et al.* on clinical isolates of *E. coli* reported no significant repercussions in bacterial motility under sub-MICs of ciprofloxacin. In contrast, other parameters such as bacterial survival, biofilm formation and curli fimbriae expression were severely compromised (Wojnicz and Tichaczek-Goska 2013). The experimental literature on the effects of NA sub-MICs is very limited. Instead, most studies have highlighted the repercussions of fluoroquinolone (ciprofloxacin and ofloxacin) exposure on bacterial motility. Nevertheless, our results are in agreement with the previous finding, which suggests that NA abrogates swarming motility in *E. coli* (Li *et al.* 1993). The molecular mechanism of NA can explain this reduction in motility. Its ability to reversibly associate with bacterial DNA causing replication arrest by virtue of inhibiting DNA gyrase and topoisomerase IV seems to be the probable route of inhibition. This fact has been substantiated by Soutourina *et al.* (2001), suggesting a strong and direct correlation between DNA supercoiling and cellular motility being influenced by environmental factors. Similarly, Gibson and colleagues confirmed the downregulation of genes regulating bacterial motility and chemotaxis in *E. coli* under the influence of a DNA gyrase inhibitor, norfloxacin (Gibson *et al.* 2006). Therefore, our findings suggest that NA at 1/2 MIC inhibits swimming and swarming motility in the commensal *E. coli* strain.

### NA at sub-MIC induces moderate biofilm formation in *E. coli*

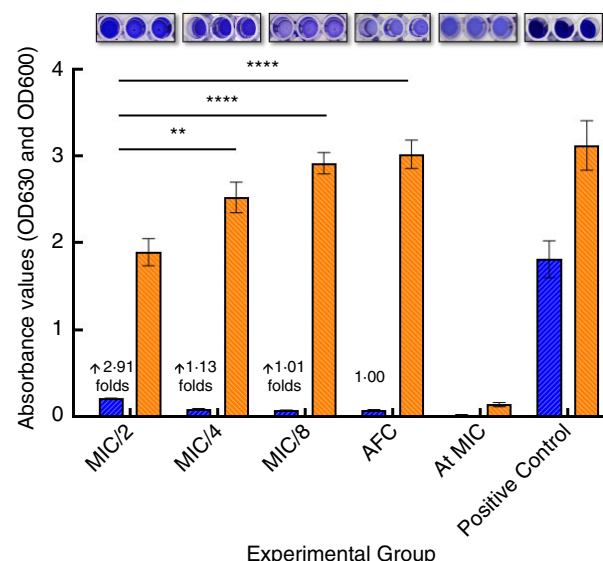
Antibiotics at sublethal levels act as cell-stimulatory molecules driving organism-wide alterations in physiology,



**Figure 2** The effect of NA sub-MICs on the motility of commensal *Escherichia coli*. (a) Representative set of images for motility assay. (b) Diameters of bacterial growth on (■) swimming and (■) swarming agar plates. Growth diameters were analysed by one-way ANOVA test and significant differences in bacterial motility were detected. The percentage reduction has been presented in relation to the untreated control (AFC). Results are represented as standard error of mean (\* $p \leq 0.1$ ).

morphology, metabolome, proteome, etc. (Pérez-Llarena and Bou 2016). This includes producing microbial biofilms that armor bacterial cells against desiccation, mechanical and shear forces, altered temperature and pH, nutrient scarcity, and even environmental stress like exposure to antimicrobial agents (Yin *et al.* 2019). Sub-MICs of various antibiotics have been shown to exert variable effects on the biofilm-forming ability in *E. coli*, including induction under amoxicillin (Chadha 2021) and inhibition with amikacin, colistin and ciprofloxacin (Dong *et al.* 2019). Nevertheless, microbial biofilms adapt and evolve rapidly in response to sub-MIC exposure, establishing a strong link with the development of AMR (Trampari *et al.* 2021). Reduction in bacterial motility upon antibiotic exposure has been previously associated with the induction of bacterial biofilms (Yildiz 2008; Chadha 2021). Since motility of commensal *E. coli* was inhibited under 1/2 MIC of NA, it prompted us to investigate the effect on biofilm formation. CV-binding assays showed that NA at 1/2 MIC stimulated the formation of a moderate biofilm in *E. coli* 48 h post-incubation, as per the previously described criteria for bacterial biofilm formation (Stepanovic *et al.* 2000). While cells grown under 1/4 and 1/8 MIC of NA failed to produce any biofilm and remained in planktonic form. The AFC also showed similar results (Fig. 3). The control *P. aeruginosa* strain

also exhibited strong biofilm formation ability. Interestingly, the cell growth under 1/2 MIC was lower than that of 1/4 and 1/8 MIC but inversely correlated with the ability to form bacterial biofilm. Taken together, our findings indicate that NA at 1/2 MIC inhibits bacterial motility and promotes biofilm formation in the commensal *E. coli* strain. Since biofilm formation in *E. coli* is an extraordinarily vast and concatenated process, the molecular mechanisms underlying this phenomenon are very complex. The development of sub-MIC-induced biofilms has been reported to coincide with the increased production of exopolysaccharides (Majtán *et al.* 2008; Chadha 2021). This involves numerous genes connected through a cascade of cellular pathways. The upregulation of positive biofilm regulators such as *bcsA*, *csgD*, *fliA*, *motA*, *pga* and *wza* genes and their activation by quorum sensors may be associated with the production of microbial biofilm upon NA exposure at sub-MIC levels (Sharma *et al.* 2016). Also, the bacterial signal transducer cyclic diguanylic acid (c-di-GMP) may be implicated with biofilm formation under NA sub-MIC. Increased cellular levels of c-di-GMP have been positively linked with the expression of bacterial matrix components such as cellulose and poly-N-acetylglucosamine, alongside repression of flagellar motility (Hall and Lee 2018). However, investigations to delineate the precise mechanism for NA-induced biofilm

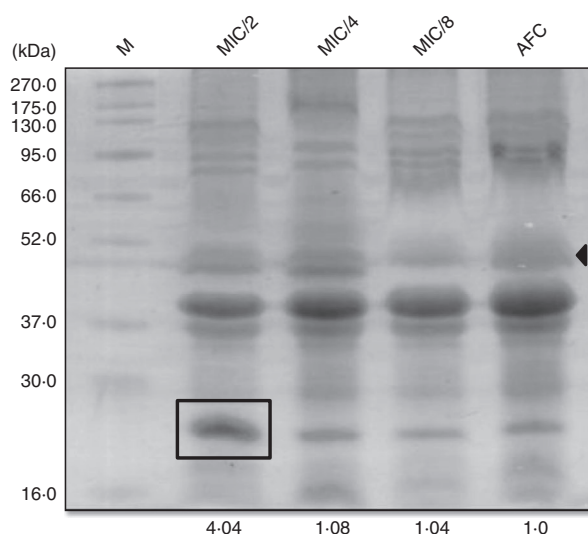


**Figure 3** Exposure to NA sub-MIC induces biofilm formation in the commensal strain of *Escherichia coli*. CV-binding assay showed moderate biofilm formation upon exposure to NA at 1/2 MIC. denotes biofilm formation (OD630) and represents cell growth (OD600). The numbers indicate the fold increase in biofilm formation in relation to AFC. Results are represented as standard error of mean (\*\* $p \leq 0.01$ , \*\*\* $p \leq 0.0001$ ). Upward arrow indicates an increase in the biofilm formation. *P. aeruginosa* was employed as a positive control.

formation in the commensal *E. coli* strain is further warranted.

#### Sub-MIC of NA drives differential OMP expression in commensal *E. coli*

Exposure to antibiotics at subinhibitory concentrations is well-known to drive differential gene expression resulting in an altered proteome in bacterial strains (Zhao *et al.* 2015; Chetri *et al.* 2019). OMPs constitute a significant proportion of the cell-bound proteins and play a critical role in bacterial survival, pathogenesis, adhesion, nutrient acquisition and translocation, etc. (Schulz 2002). Studies have implicated the role of differential OMP expression with increased cell adhesion upon amoxicillin exposure (Chadha 2021). Nevertheless, their association with nutrient uptake (as porins or siderophores) has also been established previously (Rossmanith *et al.* 1991). This intrigued us to examine the effect of NA sub-MICs on the OMP profile of the commensal *E. coli*. Interestingly, we found the OMP profiles to be differentially regulated upon NA exposure (Fig. 4). A major OMP of molecular weight  $\sim 26.12$  kDa (P26) was overexpressed by a factor of four under 1/2 MIC of NA, while its expression remained consistently at a basal



**Figure 4** SDS-PAGE analysis of OMPs isolated from NA sub-MIC treated and AFC *Escherichia coli* cells. indicates the differentially expressed major OMP upon NA exposure. signifies the variation in OMP expression. The numbers below are values of relative levels of the indicated OMP quantified and equalized to AFC lane using ImageJ software. (M) Molecular weight marker.

level under other sub-MICs as well as in the AFC. Minor variation in OMP expression was observed in 1/8 MIC and AFC corresponding to molecular weight protein bands  $\sim 50.06$  kDa and  $47.14$  kDa. Other major and minor bands of OMPs were expressed uniformly under all experimental conditions, regardless of the antibiotic treatment at sub-MIC levels. Therefore, our results suggest that NA at a concentration of 1/2 MIC regulates differential OMP expression in the commensal *E. coli* strain. This may be related to a differential gene expression triggered by NA at 1/2 MIC. Considering the involvement of OMPs in bacterial virulence and nutrient uptake (Rossmanith *et al.* 1991; McKenney *et al.* 1994; Vaca *et al.* 2020), we speculated P26 to be either an iron-quenching protein or a cell-adhesion molecule. However, Chrome Azure S (CAS) agar assays (Fig. S2) and *in vitro* cell-adhesion assays with HT-29 cells (Fig. S3) negated our hypothesis. Nevertheless, previous accounts on different bacterial OMPs suggest their role in imparting resistance against antimicrobial peptides, bile salts and serum (Lin *et al.* 2002). The upregulation of P26 may therefore be inferred to increased resistance of *E. coli* towards antimicrobial agents. Nevertheless, these speculations can be precisely substantiated by characterization of P26 using MALDI-TOF and elucidating its role by bench-based studies. Our findings ultimately point to the plausible fact that sub-inhibitory concentrations of antibiotics act as elicitors that drive differential gene expression in bacteria.

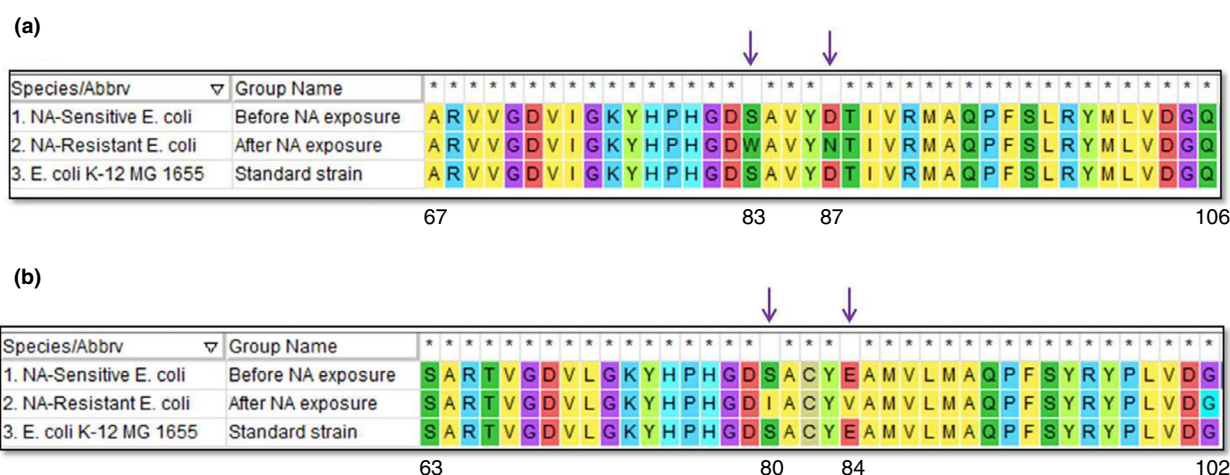


### Passaging under sub-MIC of NA induces drug-mediated resistance in commensal *E. coli*

This study was initially undertaken to examine the effects of NA sub-MICs on bacterial responses and physiology in a commensal *E. coli* strain. Since our results identified significant changes in bacterial physiology, we further planned to investigate the role of long-term exposure of NA sub-MICs on the commensal strain. During regular passaging under sub-MICs of NA for extended time, we observed prolific growth in cultures supplemented with NA at 1/2 MIC only after 24 days. The growth profile and  $A_{600}$  values matched with that of the AFC till 12 h, raising speculations of bacterial contamination, antibiotic degradation/inactivation, or the least possible chance of antibiotic (NA) resistance. Possibilities of bacterial contamination and NA inactivation were nullified following patch-plating of suspected culture onto Endo agar plates and the use of freshly prepared NA stock solutions, respectively (data not shown). The least probable outcome, that is, development of NA resistance, was then probed into using the microbroth dilution method under test concentrations of NA ranging between 1 and  $512 \mu\text{g ml}^{-1}$ . To our surprise, the MIC of this subcultured strain increased from  $6 \mu\text{g ml}^{-1}$  (sensitive; parent strain) to  $64 \mu\text{g ml}^{-1}$  (resistant; mutant strain) following repeated exposure to NA at 1/2 MIC. To rule out the possibility of this resistant strain being genotypically different from the parental type, genomic DNA was isolated from both cultures, followed by amplification of 16S rDNA and its double-pass sequencing. Alignment of 16S rDNA revealed the resistant strain to be identical to that

of the parent strain. This confirmed the development of resistance in the commensal *E. coli* strain following NA exposure at 1/2 MIC.

To delineate the cause of evolved NA resistance in *E. coli*, the QRDRs of DNA gyrase (*gyrA*, *gyrB*) and topoisomerase IV (*parC*, *parE*) were PCR amplified from the genomic DNA of parental (NA sensitive) and mutant (NA resistant) strains. Multiple sequence alignment of the translated QRDR sequences revealed amino acid substitutions in the *gyrA* and *parC* proteins (Fig. 5). No amino acid variations were observed in the QRDRs of *gyrB* and *parE* (Fig. S4). The mutant strain shared identical amino acid residues for the latter QRDRs with the parental strain. The mutations identified in the *gyrA* gene include Ser83Trp (S83W) and Asp87Asn (D87N). These *gyrA* mutations have been characterized and associated with quinolone and fluoroquinolone resistance in uropathogenic *E. coli* and *Shigella* isolates (Anjum *et al.* 2018; Yu *et al.* 2020). The presence of these amino acid variations in *gyrA* has been shown to reduce the binding affinity of the drug to the enzyme-DNA complex (Hooper and Jacoby 2015). The precise molecular mechanism presented for this mutation is the induction of a local conformational change (hydrophilic to hydrophobic residue) in the gyrase A subunit resulting in complete loss of enzyme-drug interaction (Pourahmad Jaktaji and Mohiti 2010). Similarly, Ser80Ile (S80I) and Glu84Val (E84V) mutations were identified in the QRDR of *parC*. The S80I mutation responsible for imparting fluoroquinolone resistance in *E. coli* has been previously identified (Minarini and Darini 2012). The E84V variation in *parC* QRDR has been recognized as one of the most critical mutations responsible



**Figure 5** Multiple sequence alignment of translated amino acid sequences of the QRDRs of (a) *gyrA* and (b) *parC* genes in the commensal *Escherichia coli* strain. Sequences corresponding to the QRDRs of NA-sensitive *E. coli* and *E. coli* K-12 MG1655 *gyrA* and *parC* were employed as control and reference, respectively. Amino acid variations from the wild-type sequence exhibiting resistance to NA are marked with arrows (↴).

for quinolone resistance in *E. coli* (Ardebili *et al.* 2015). Interestingly, an association between simultaneous mutations in the QRDRs of *gyrA* and *parC* has been reported to play a major role in high-level fluoroquinolone resistance (Ardebili *et al.* 2015). The overall outcome of these amino acid variations appears to cumulatively impede the binding of NA to its cellular target, thereby resulting in the emergence of antibiotic resistance. Our findings are in accordance with the previously characterized mutations in the QRDRs of DNA gyrase and topoisomerase IV genes. This strongly points to the fact that antibiotics at sub-MICs act as stimulatory molecules that can solely induce anthropogenic resistance in the environment.

All our experimental findings were reported at 1/2 MIC of NA, but not at other sub-MICs. The most appropriate justification for this phenomenon is that antibiotics regulate gene expression and physiological responses in a dose-dependent manner. Therefore, it can be inferred that antibiotic-mediated bacterial responses are concentration-dependent and demonstrate the fundamental potential to drive the emergence of anthropogenic resistance. The induction of a bacterial biofilm accompanied by the emergence of DNA gyrase and topoisomerase mutants under the influence of NA sub-MIC exemplifies how antibiotics circumvent the transition of a commensal strain to a virulent type. This may lead to the dissemination of antibiotic resistance among the intestinal microflora and, ultimately, the non-intestinal pathogens. Therefore, the consequences of antibiotics at sub-lethal levels over the commensal microflora should not be undermined. Nevertheless, *in vitro* (experimental) and *in vivo* (physiological) growth conditions vary drastically in terms of nutrients, microenvironment, pH, oxygen content and microbial diversity. Furthermore, limiting space for gut colonization, intestinal dysbiosis, and the rate of antibiotic metabolism are some factors that can influence bench-based studies. The molecular dynamics of bacterial motility, biofilm formation and OMP expression are pretty complex in nature. These involve multifaceted proteins and intricate signalling pathways that can be discretely identified using microarray-based methods or RNA sequencing following NA exposure. This will help in understanding the gene regulation by NA *in vitro*. Additionally, whole-genome sequencing of the commensal strain before and after 7 days of passing under NA sub-MICs can help decipher other new mutations responsible for imparting drug resistance, apart from the ones reported in the QRDRs. Once these findings are validated, experimentation with animal models can help decipher the role of NA in shaping microbial responses and their clinical outcomes. Overall, these experimentations can help in devising antibiotic regimens for optimal antimicrobial therapy without incurring the risk of drug resistance.

In summary, the present study unravels the intriguing possibilities regarding the invariable effects imposed by

NA sub-MICs on a commensal strain of *E. coli*. The fact that NA sub-MICs can significantly reduce bacterial motility, induce biofilm formation and regulate differential OMP expression suggests their ability to impact the physiology and functioning of commensal bacteria. Additionally, the emergence of mutations in the QRDRs demonstrates that NA at sub-MIC levels can play a crucial role in inducing anthropogenic resistance. A better understanding in this regard may help in the establishment of a relationship between the repercussions brought by antibiotic sub-MICs on commensal microbiota for optimizing antimicrobial therapy.

## Materials and methods

### Bacterial strains, media and growth conditions

A commensal strain of *E. coli* (Phylogroup A, GenBank accession no. MT305955.1) isolated from River Yamuna, New Delhi, India was used in this study. The strain was cultured in Luria Bertani (LB) broth (HiMedia Labs, Mumbai, India), unless otherwise indicated. The MIC value of NA for the commensal *E. coli* strain was determined to be  $6.0 \mu\text{g ml}^{-1}$  (Fig. S1) using Nalidixic acid Ezy MIC<sup>TM</sup> strips (HiMedia Labs). For all experimental procedures, the strain was passaged in LB supplemented with NA at sub-MIC levels: 1/2 MIC ( $3.0 \mu\text{g ml}^{-1}$ ), 1/4 MIC ( $1.5 \mu\text{g ml}^{-1}$ ) and 1/8 MIC ( $0.75 \mu\text{g ml}^{-1}$ ) for 7 days at  $37^\circ\text{C}$  with mild shaking. The culture obtained on the eighth day was used for further investigations. Cells grown in the absence of NA served as positive growth control, while those supplemented with NA at MIC were employed as negative growth control. As significant changes in bacterial physiology were observed upon NA treatment, we were intrigued to examine the effects over long-term NA exposure. Hence, the strain was regularly passaged under NA sub-MICs alongside the experimental setup. *E. coli* ATCC 25922 was used as quality control. *Pseudomonas aeruginosa* was employed in crystal violet (CV) assays as a biofilm-forming control.

### Bacterial growth kinetics

Sub-MIC exposed bacterial culture was inoculated (1%) in LB medium containing the indicated NA sub-MICs and incubated at  $37^\circ\text{C}$  with shaking at  $200 \text{ rev min}^{-1}$ . Aliquots were collected every 2 h from the growing cultures, and the growth rate was estimated by measuring the changes in absorbance 600 nm over a 12 h span. Antibiotic-free culture (AFC) was employed as positive growth control, while NA supplemented at MIC was used as negative growth control. *E. coli* ATCC 25922 served as quality control for the experiment.

### Bacterial motility assays

To examine the effect on bacterial motility, sub-MIC treated cultures were spot inoculated on swimming (LB, 0.3% agar) and swarming (LB, 0.6% agar) media plates supplemented with sublethal concentrations of NA, as described previously (Walsh *et al.* 2019). The diameters of bacterial growth were measured using a Vernier caliper. Culture plates lacking NA served as positive growth controls, while those containing NA at MIC were employed as negative growth controls.

### Static biofilm formation assay

Biofilm formation under static conditions was examined using CV assay in a 96 well flat-bottom microtiter plate (Tarsons, Kolkata, India) (Chadha 2021). Sub-MIC-treated cultures corresponding to  $A_{600}$  value of 0.5 McFarland standards (containing  $1 \times 10^8$  cells per ml) were dispensed into LB-containing wells supplemented with NA at mentioned sub-MIC levels. Six wells were designated for each experimental group. Cultures were grown in a static incubator at 37°C for 48 h. Post-incubation,  $A_{600}$  was recorded for all cultures to determine bacterial growth, and the supernatants were carefully removed by gentle pipetting. Wells were washed thrice with PBS to remove any unadhered bacteria. Biofilms were then stained using 0.1% CV at room temperature (RT) for 20 min. After staining, the dye was removed, and wells were washed thrice with PBS to remove excess CV. The plate was air-dried at 37°C, followed by the addition of 33% glacial acetic acid to each well to elute the bound dye. Absorbance values were then measured using an ELISA plate reader (Bio-Rad, Hercules, CA) at 630 nm. Wells containing only media (LB) were used for blanking. Those with cells without NA served as positive growth controls, while those with NA at MIC were used as negative growth controls. Biofilm forming abilities were classified into negative, weak, moderate or strong based upon the  $A_{630}$  values of bacterial biofilms (Stepanovic *et al.* 2000). We defined a cut-off value ( $A_{CO}$ ) by evaluating the mean  $A_{630}$  of negative controls. The following mentions the criteria: (1)  $A_{630} < A_{CO}$ : negative for biofilm formation, (2)  $A_{630} > A_{CO}$ : weak biofilm formation, (3)  $4 \times A_{CO} > A_{630} > 2 \times A_{CO}$ : moderate biofilm formation, and (4)  $A_{630} > 4 \times A_{CO}$ : strong biofilm-forming ability. Biofilm-forming *P. aeruginosa* was also employed as a positive control.

### Isolation of outer membrane proteins

Outer membrane proteins (OMPs) were isolated from *E. coli* cells using a modified protocol (Reeks *et al.* 2005).

50 ml of sub-MIC exposed cultures and AFC were pelleted by centrifugation at 12 000 g. Cell pellets were washed twice with 10 m mol l<sup>-1</sup> 4-(2-Hydroxyethyl)-1-piperazineethanesulfonic (HEPES) buffer (pH 7.4) and then resuspended in 10 ml HEPES buffer containing 2 m mol l<sup>-1</sup> PMSF and DTT. Cell suspensions were sonicated on ice at 25 MHz for 6–8 cycles with 15 s sonication and 45 s off-pulse. The lysates were centrifuged at 2000 g for 30 min at 4°C. Supernatants were then transferred to ultracentrifuge tubes and centrifuged at 100 000 g for 60 min at 4°C (Himac CS150NX, Hitachi, Tokyo, Japan). Following centrifugation, the supernatants were discarded, and the pellets resuspended in 1 ml of 10 m mol l<sup>-1</sup> HEPES buffer and 1 ml of 2% sodium-N-lauroyl sarcosinate (Sigma, St. Louis, MO). Samples were incubated at RT for 30 min, followed by another round of ultracentrifugation. Supernatants were discarded, and the protein pellets were resuspended in nuclease-free water (Invitrogen, Carlsbad, CA) containing PMSF and DTT at 2 m mol l<sup>-1</sup> each. Protein concentrations were estimated by Bradford's assay using Bio-Rad Protein Assay Dye Reagent Concentrate (Bio-Rad Laboratories, Hercules, CA). OMP samples (80 µg each) were resolved by SDS-PAGE on a 12% resolving gel, alongside pre-stained protein ladder as molecular size standards. The relative protein concentrations of differentially expressed OMPs were quantified by densitometric analysis and equalized to OMPs in AFC using ImageJ software, where bovine serum albumin (BSA) served as a standard.

### Assessment of amino acid substitutions in the QRDRs of sub-MIC treated *E. coli*

To delineate the cause of NA resistance in the commensal strain, genomic DNA was isolated from NA-resistant and -sensitive *E. coli* cells by boiling lysis. PCR amplification of the QRDRs of chromosomal genes related to quinolone resistance *viz.* *gyrA*, *gyrB*, *parC* and *parE* were carried out using primers described in Table S1. The amplicon size was confirmed, and the absence of non-specific amplicons was determined by electrophoresing the samples in 1–2% agarose gel. Single-pass sequencing of the amplicons was performed at Central Instrumentation Facility, UDSC (New Delhi, India). The sequences were translated in the same frame as the complete coding DNA sequence of the corresponding genes available from NCBI GenBank using the translate tool available at ExPASy (<http://web.expasy.org/tools/translate/>). Variations in the amino acid sequences of QRDRs of NA-resistant *E. coli* were determined by sequence alignment with QRDRs of corresponding NA-sensitive *E. coli* and wild-type sequences of *E. coli* strain K-12 MG 1655 (NCBI GenBank accession number U00096) using ClustalW program (<http://www.ebi.ac.uk>).



## Statistical analysis

Each assay was performed in triplicates, and the mean values  $\pm$  standard deviations were calculated. Experiments were conducted in three independent events, and a representative set of data has been reported. The significance of experimental data were evaluated with a one-way ANOVA test using GraphPad Prism ver. 8.0. *p*-values of  $<0.05$  were considered to be statistically significant (*p* values: \*  $\leq 0.05$ , \*\*  $\leq 0.01$  \*\*\*\*  $\leq 0.0001$ ).

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## Conflict of Interest

No conflict of interest declared.

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## Supporting Information

Additional Supporting Information may be found in the online version of this article:

**Figure S1** Determination of NA MIC against *E. coli* using E-test (black arrow indicates the MIC).

**Figure S2** Siderophore production assay for sub-MIC treated cultures using CAS agar. (NA/2) 1/2 MIC, (NA/4) 1/4 MIC, (+ve) AFC.

**Figure S3** Cell adhesion experiments with sub-MIC treated *E. coli* cells to HT-29 cells *in vitro*. Adhesion scores are represented as the average number of *E. coli* cells binding per HT-29 cell. Results are represented as standard error of mean.

**Figure S4** Multiple sequence alignment of translated amino acid sequences of the QRDRs of (A) *gyrB* and (B) *parE* genes in the commensal *E. coli* strain. Sequences corresponding to the QRDRs of NA-sensitive *E. coli*, and *E. coli* K-12 MG1655 *gyrB* and *parE* were employed as control and reference, respectively.

**Table S1** Details of primers used for analysis of QRDR of chromosomal genes.

**Table S2** Comparison of growth rate and doubling time of *E. coli* under sub-MICs of NA.